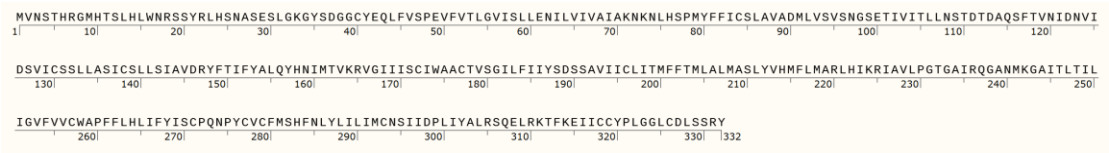
1. Choose the HUMAN obesity genes MC4R (Melanocortin 4 receptor), use homologene (https://www.ncbi.nlm.nih.gov/homologene) to find its homolog sequence across at least three species,

Show the aa sequence alignment for homologs you found;

Explore your gene from UCSC genome server and download a 1 kb promoter region (-1000 - -1).
Helps:
http://genome.ucsc.edu/goldenPath/help/hgTracksHelp.html;

AA sequence alignment:

Homo sapiens:



C. lupus:



B. taurus:



1 kb promoter region:

Homo sapiens:

agcagctacaaactatagagcactaaaaccaaaaataatcatagaagaaa
agaatatacaaaatttttgtataatctgttgttcagttaatttggaaagta
actttgctgccatctttcaaaccaccttaagcaaaatattcagaaacttc
aataactaactagtatcaccagtcttggcttgataagtaaagttactagg
tatttgtcggttcaacttacgatacgttaaactctggaggtttaaaaaaa
agttgcttaaaaaaaaaaaacagaatgcagcttattattttcctgagttc
agtgtaagcatagttacattaatgccgtatgagtttaaaatattgtaaat
aaaaagcacctttttttcagctataacacaaactcctcaatagagaaatat
gggatattagtgcattaaaatcaacccctttttgaacccattcataattta
tagctacctacatagttcttccagccataccatgtctatctgtatcatta
aatatcctctctgtttttcaggtattttaactgaaccactactggctggg
ttcttaataaaaacaaaaaattaagaaggtaaataaattcagtatcctag
ccaaaaagcagagtatctgtttcaggaacataagacagtatactttccat

ttcattttacacacacacatgatgtcttttacataagaatgaacctggct
gcctgaagatagcttgttttgatatcacagaatggcctctagggatcaag
ttgtttcccttgaaaacttaaaaagggaaattcagtctcttatccggctt
gcttctcgacttcagacttcaggagcttctgcatctgaatctgtgtcacc
ctgcaatttgtgtctctgaaacataatcctgtctcacagactcccattgc
attgggattggtcagaaggaagcagaggaggagccaccgtgaacatgttt
tttttctctccacacatcataaaaatcagcagctggcgcctacagcccct

2. You have the following sequence reads from a genomic clone of the *Homo sapiens* genome:

Read 1: ATGCGATCTGTGAGCCGAGTCTTTA

Read 2: AACAAAAATGTTGTTATTTTTATTTCAGATG

Read 3: TTCAGATGCGATCTGTGAGCCGAG

Read 4: TGTCTGCCATTCTTAAAAACAAAAATGT

Read 5: TGTTATTTTTATTTCAGATGCGA

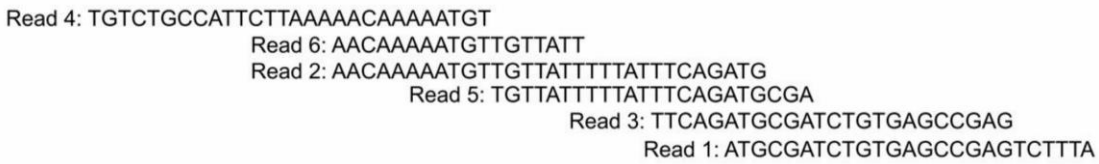
Read 6: AACAAAAATGTTGTTATT

a. Use these six sequence reads to create a sequence contig of this part of the *H. sapiens* genome.

b. Translate the sequence contig in all possible reading frames.

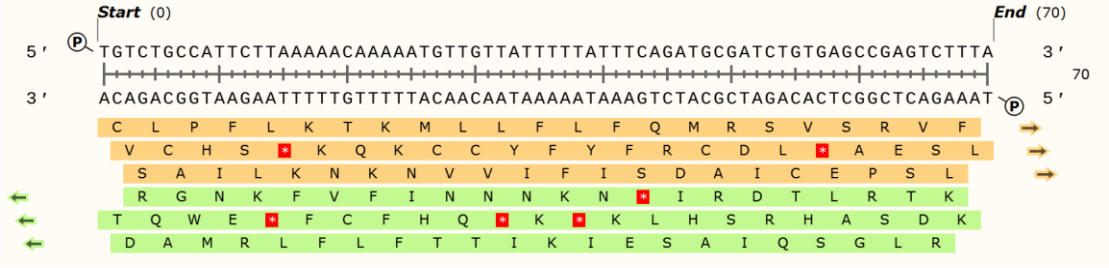
c. Go to the BLAST page (http://www.ncbi.nlm.nih.gov/BLAST/) and see if you can identify the gene of which this sequence is a part by using each of the reading frames as a query for protein–protein comparison (BLASTp).

A．拼接后得到：



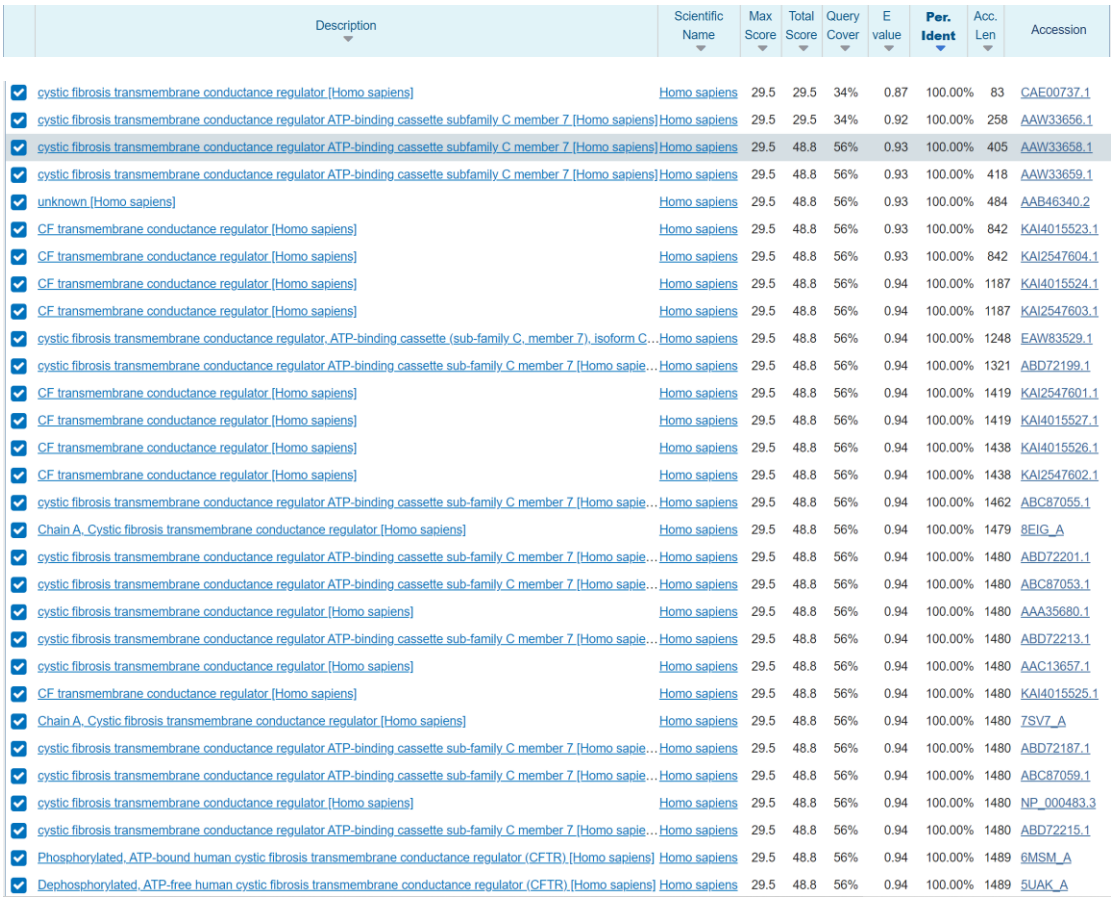
TGTCTGCCATTCTTAAAAACAAAAATGTTGTTATTTTTATTTCAGATGCGATCTGTGAGCCGAGTCTTTA

B．将所得序列选择 ORFs 后转换成可能的蛋白质序列如下：

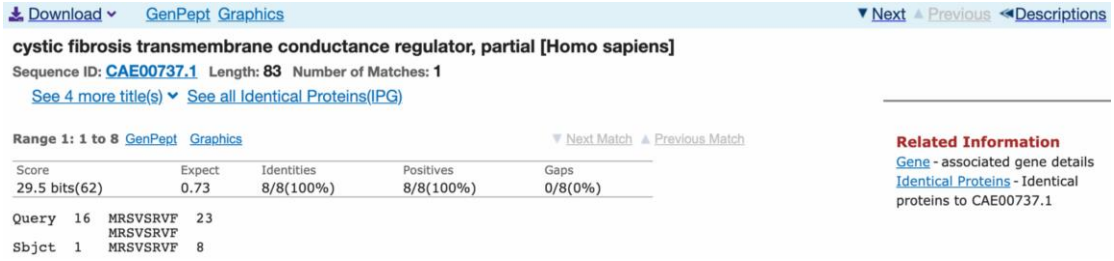


可能的蛋白序列（相似程度 100%，E value 小于 1）有：

1. CLPFLKTKMLLFLFQMRSVSRVF



Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
cystic fibrosis transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	29.5	34%	0.87	100.00%	83	CAE00737.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette subfamily C member 7 [Homo sapiens]	Homo sapiens	29.5	29.5	34%	0.92	100.00%	258	AAW33656.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette subfamily C member 7 [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.93	100.00%	405	AAW33658.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette subfamily C member 7 [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.93	100.00%	418	AAW33659.1
unknown [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.93	100.00%	484	AAB46340.2
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.93	100.00%	842	KAI4015523.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.93	100.00%	842	KAI2547604.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1187	KAI4015524.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1187	KAI2547603.1
cystic fibrosis transmembrane conductance regulator, ATP-binding cassette (sub-family C, member 7), isoform C...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1248	EAW83529.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1321	ABD72199.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1419	KAI2547601.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1419	KAI4015527.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1438	KAI4015526.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1438	KAI2547602.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1462	ABC87055.1
Chain A, Cystic fibrosis transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1479	8EIG_A
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	ABD72201.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	ABC87053.1
cystic fibrosis transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	AAA35680.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	ABD72213.1
cystic fibrosis transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	AAC13657.1
CF transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	KAI4015525.1
Chain A, Cystic fibrosis transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	7SV7_A
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	ABD72187.1
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	ABC87059.1
cystic fibrosis transmembrane conductance regulator [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	NP_000483.3
cystic fibrosis transmembrane conductance regulator ATP-binding cassette sub-family C member 7 [Homo sapie...	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1480	ABD72215.1
Phosphorylated, ATP-bound human cystic fibrosis transmembrane conductance regulator (CFTR) [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1489	6MSM_A
Dephosphorylated, ATP-free human cystic fibrosis transmembrane conductance regulator (CFTR) [Homo sapiens]	Homo sapiens	29.5	48.8	56%	0.94	100.00%	1489	5UAK_A



Download GenPept Graphics ▼Next ▲Previous ◀Descriptions

cystic fibrosis transmembrane conductance regulator, partial [Homo sapiens]

Sequence ID: CAE00737.1 Length: 83 Number of Matches: 1

See 4 more title(s) See all Identical Proteins(IPG)

Range 1: 1 to 8 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Identities	Positives	Gaps
29.5 bits(62)	0.73	8/8(100%)	8/8(100%)	0/8(0%)

```
Query  16  MRSVSRVF  23
           MRSVSRVF
Sbjct  1   MRSVSRVF  8
```

Related Information
Gene - associated gene details
Identical Proteins - Identical proteins to CAE00737.1

2. VCHS

   没有 E value 小于 1 的结果

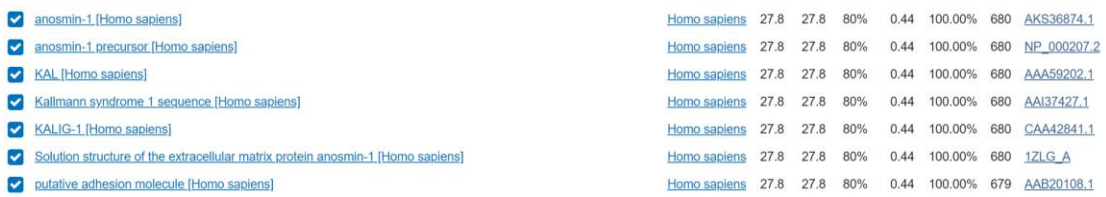
3. SAILKNKNVVIFISDAICEPSL

   没有 E value 小于 1 的结果

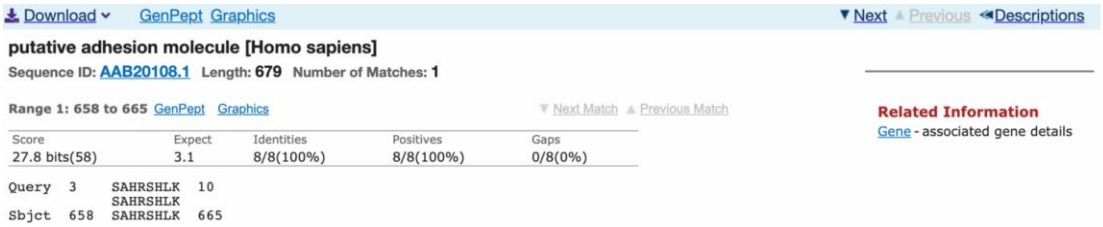
4. KTRLTDRI

   没有 E value 小于 1 的结果

5. KDSAHRSHLK



anosmin-1 [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	680	AKS36874.1
anosmin-1 precursor [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	680	NP_000207.2
KAL [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	680	AAA59202.1
Kallmann syndrome 1 sequence [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	680	AAI37427.1
KALIG-1 [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	680	CAA42841.1
Solution structure of the extracellular matrix protein anosmin-1 [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	680	1ZLG_A
putative adhesion molecule [Homo sapiens]	Homo sapiens	27.8	27.8	80%	0.44	100.00%	679	AAB20108.1



6. MAD（以 ATG 作为起始位置）

No significant similarity found.

综合 identity，Score 和 E-value，认为 CLPFLKTKMLLFLFQMRSVSRVF 这一段是编码人源基因 cystic fibrosis transmembrane conductance regulator, partial [Homo sapiens]。